

Ali Rizvi

Senior Data Engineer

✉ ali.rizvi6627@gmail.com 📍 Bartlett IL 60103

PROFILE

Data Engineer with over eight years of experience building scalable, reliable data platforms that support critical business decisions. I have designed and maintained large-scale batch and real-time pipelines using Spark, Kafka, and Airflow across AWS, Azure, and GCP, processing multi-terabyte datasets in production environments. I have led cloud modernization efforts that improved data availability, strengthened governance, and reduced infrastructure overhead. My work has enabled analytics, reporting, and machine learning teams to deliver faster insights and drive meaningful operational and strategic impact.

PROFESSIONAL EXPERIENCE

Senior Data Engineer, Addepto

03/2023 – Present

- Architected and implemented a cloud-native **Lakehouse platform (AWS S3 + Spark 3.x + Delta Lake + Glue Catalog)** processing multi-terabyte daily batch and streaming workloads with 99.9% SLA reliability.
- Designed distributed ETL/ELT pipelines using **PySpark and Airflow**, implementing CDC (Debezium/Kafka), watermarking, and incremental merge strategies to reduce processing time by strong margin
- Engineered high-throughput real-time streaming pipelines using **Kafka + Spark Structured Streaming**, achieving exactly-once semantics, checkpointing, and sub-second latency for 3M+ events/minute.
- Optimized Spark workloads using AQE, broadcast joins, skew handling, partition pruning, Z-order clustering, and Kryo serialization, reducing shuffle spill and compute cost by 43%
- Led cloud migration from on-prem Hadoop (HDFS/Hive) to AWS EMR + S3, implementing Terraform-based IaC, IAM security models, and CI/CD automation for data pipelines.
- Implemented enterprise data modeling (Star Schema, SCD Type 2, Data Vault), improving BI query latency by 60% in Snowflake/Redshift.

Data Engineer, Binariks

07/2020 – 02/2023

- Developed large-scale distributed ETL pipelines using **Spark (Core/SQL) on Hadoop YARN**, processing 5TB+ daily batch data with optimized partitioning and bucketing strategies.
- Built Kafka-based ingestion pipelines with scalable topic partitioning and consumer group balancing for near-real-time analytics integration.
- Designed dimensional data warehouse models (Fact/Dimension, SCD Type 1/2) and implemented incremental loading strategies using high-watermark logic.
- Optimized Hive and Spark SQL queries via execution plan analysis, join reordering, partition pruning, and vectorized execution, reducing long-running queries by 67%.
- Developed and maintained 90+ production Airflow DAGs with SLA monitoring, retry logic, dynamic task generation, and automated alerting.
- Implemented data validation and reconciliation frameworks to improve pipeline accuracy and reduce downstream reporting discrepancies.

Junior Data Engineer, OSI Engineering

02/2018 – 05/2020

- Developed SQL-based ETL workflows and batch ingestion scripts integrating relational databases into Hadoop data lake environments.
- Created Hive external/managed tables with partitioning strategies to support optimized analytical queries.
- Assisted in Spark job development and debugging by analyzing executor logs, optimizing memory allocation, and resolving shuffle bottlenecks.
- Automated ingestion and monitoring processes using shell scripting and cron scheduling.
- Supported Hadoop cluster operations including HDFS data balancing, metadata management, and job performance troubleshooting.
- Contributed to migration of legacy ETL processes to distributed Spark-based architecture.

CERTIFICATES

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| • Cloudera Certified Associate (CCA) Data Analyst | • AWS Certified Big Data | • Databricks Certified Associate Developer for Apache Spark |
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SKILLS

Programming & Software Engineering

Advanced Python, PySpark, Scala, SQL (ANSI SQL, T-SQL, PL/SQL), Java, Bash, Shell Scripting; OOP, Functional Programming, Data Structures & Algorithms, Multithreading, REST API development, Unit Testing (PyTest, ScalaTest), Code modularization, Design Patterns (Factory, Singleton, Builder), Git (GitHub, GitLab, Bitbucket).

Big Data & Distributed Computing

Apache Spark (Core, SQL, Structured Streaming), Spark Performance Optimization (AQE, Catalyst, Tungsten, Shuffle Optimization, Broadcast Joins), Hadoop (HDFS, YARN, MapReduce, Hive, Pig), Presto, Trino, File Formats (Parquet, ORC, Avro, Delta Lake), Data Compression (Snappy, Gzip, LZO).

Cloud Data Platforms

AWS: S3, EMR, Glue, Redshift, Athena, Lambda, Kinesis, IAM, VPC, CloudWatch. Azure: ADLS, Data Factory, Databricks, Synapse, Event Hub. GCP: BigQuery, Dataflow, Pub/Sub, Cloud Storage, Dataproc.

Streaming & Real-Time Data Engineering

Apache Kafka (Producers, Consumers, Streams API), Kafka Partitioning & Replication, Exactly-Once Semantics, Spark Structured Streaming, Apache Flink, AWS Kinesis, GCP Pub/Sub, Event-Driven Architecture, Real-Time Monitoring.

Data Warehousing & Modeling

Dimensional Modeling (Star, Snowflake Schema), Data Vault 2.0, Slowly Changing Dimensions, Surrogate Key Strategies, Aggregation Tables, OLAP, Query Optimization, RDBMS (PostgreSQL, MySQL, SQL Server, Oracle), Cloud Data Warehouses (Snowflake, Redshift, BigQuery), NoSQL (MongoDB, Cassandra, DynamoDB).

ETL/ELT & Data Pipeline Engineering

Batch & Micro-Batch Processing, CDC, Debezium, Metadata-Driven Pipelines, Apache Airflow DAGs, Dynamic DAGs, SLA Monitoring.

DevOps & Infrastructure Engineering

Docker, Kubernetes (Spark), Terraform, CI/CD (Jenkins, GitHub Actions, GitLab CI).

PROJECTS

Scalable Data Pipeline for IoT Sensors Using Apache Flink & Kafka,

Technologies: Apache Flink, Apache Kafka, Apache HBase, REST APIs

Developed a scalable data pipeline to ingest data from thousands of IoT sensors deployed in the field. Apache Kafka was used for ingesting real-time sensor data, while Apache Flink processed the data with event-time windows and complex stream transformations. The processed data was stored in HBase for further analysis and real-time visualization through a REST API.

ETL Automation Using Apache Airflow for Healthcare Data Integration,

Technologies: Apache Airflow, PostgreSQL, AWS S3, Python

Built an automated ETL pipeline using Apache Airflow to extract healthcare data from various sources (FHIR API, CSV, and XML), transform it into a structured format, and load it into a centralized data warehouse on PostgreSQL. The pipeline was designed with dynamic DAGs to handle different data sources and scheduling needs, including retry logic, SLA monitoring, and data quality checks.

Cloud-Native Data Warehouse Design with Snowflake and Fivetran,

Technologies: Snowflake, Fivetran, AWS S3, SQL

Architected and implemented a cloud-native data warehouse solution using Snowflake and Fivetran. Utilized Fivetran for automated data extraction and integration from multiple sources, including marketing, sales, and customer service platforms. Designed Snowflake's multi-cluster architecture to handle high concurrency and data transformation workloads, significantly improving query performance.

EDUCATION

Bachelors of Science